DC to DC Converters

Difference between ‘Chopper’ and ‘Converter’

BUCK CHOPPER

$$V_o = \frac{t_{on}}{t_{on} + t_{off}} V_s$$

$$\Rightarrow V_o = \alpha V_s$$

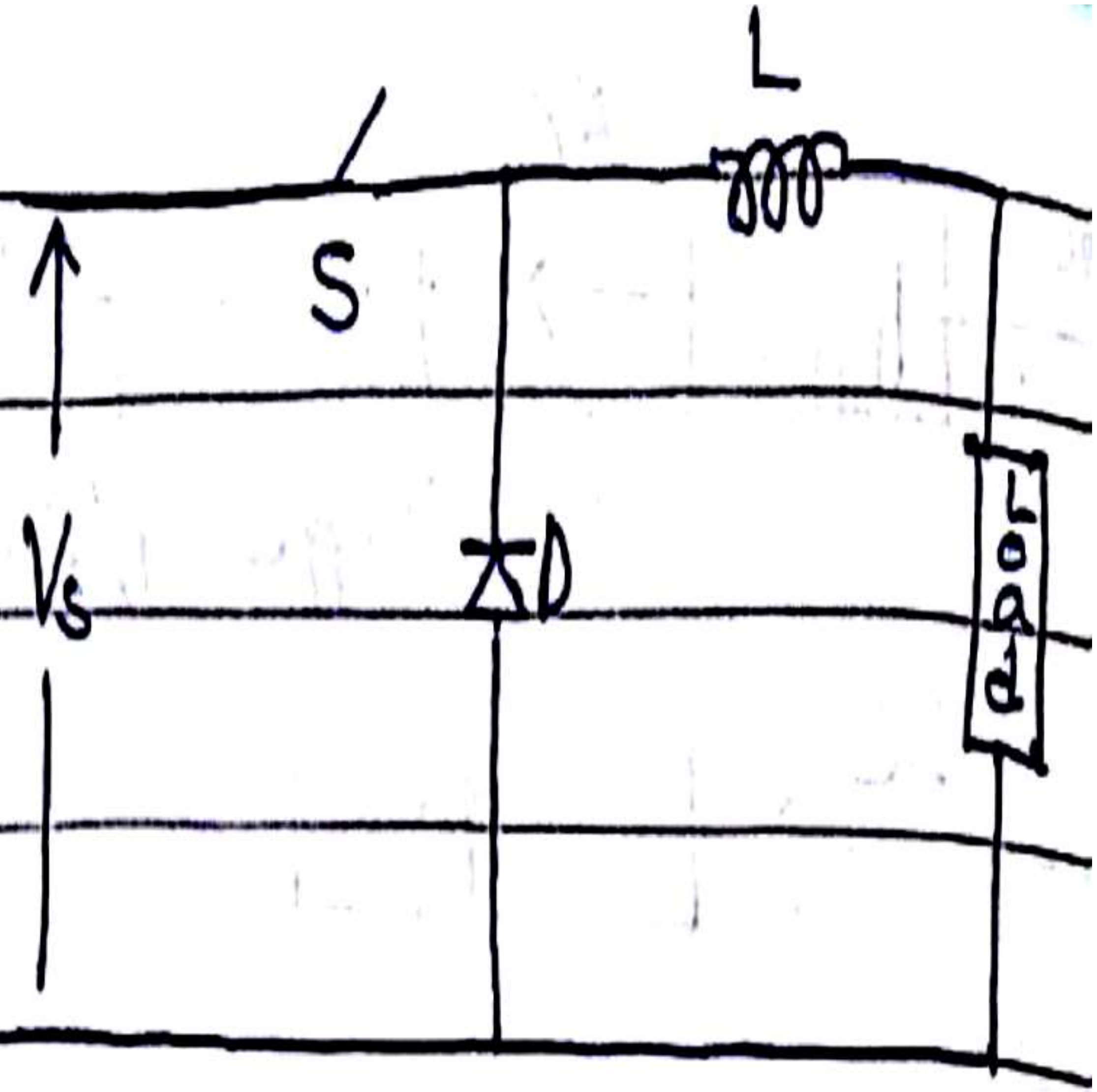


Fig. 17.1(a): Buck converter (dc-dc)

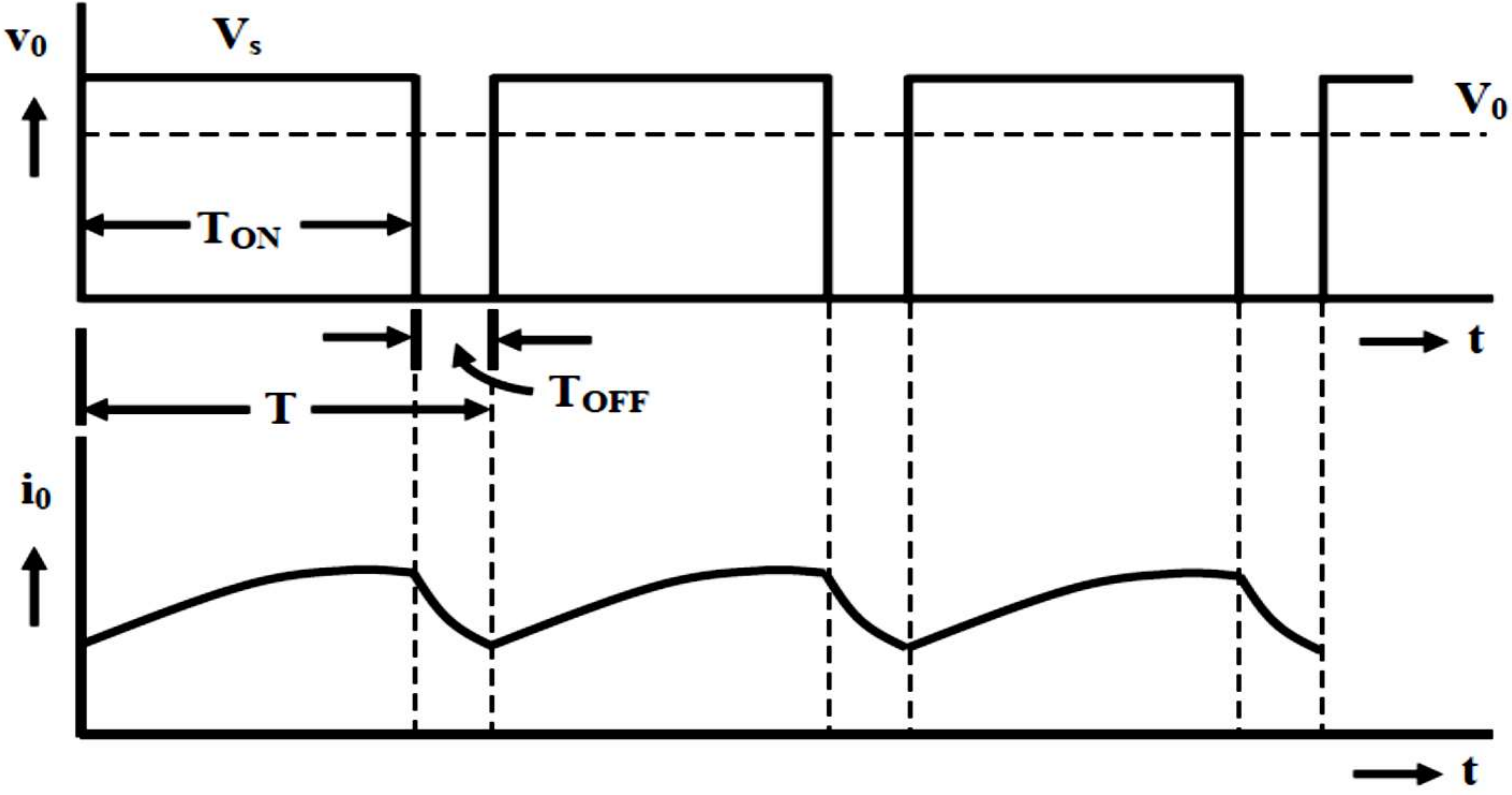
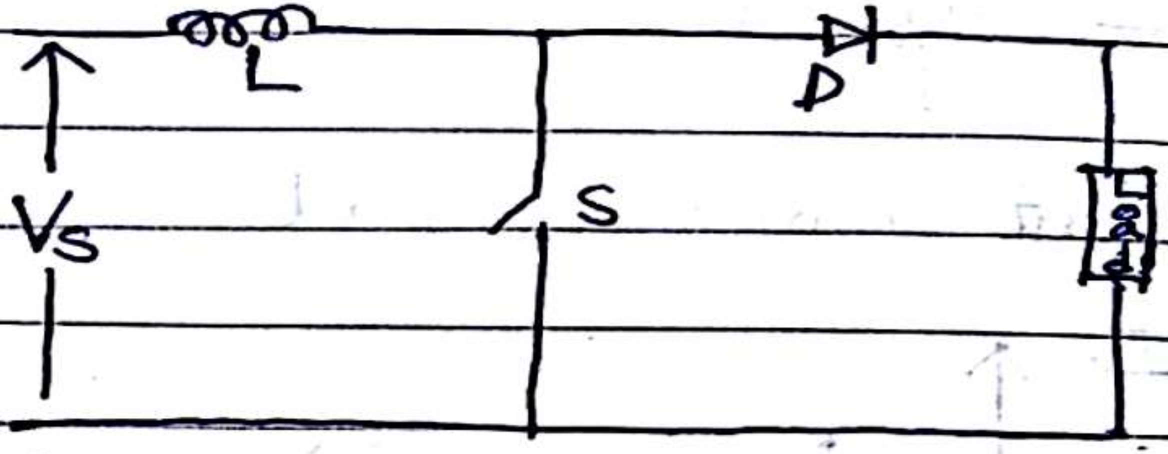


Fig. 17.1(b): Output voltage and current waveforms

BOOST CHOPPER



When 'S' is ON

$$v_L = V_s$$

When 'S' is OFF

$$V_s + v_L - v_o = 0$$

$$\Rightarrow v_L = v_o - V_s$$

$$V_s T_{on} = (v_o - V_s) T_{off}$$

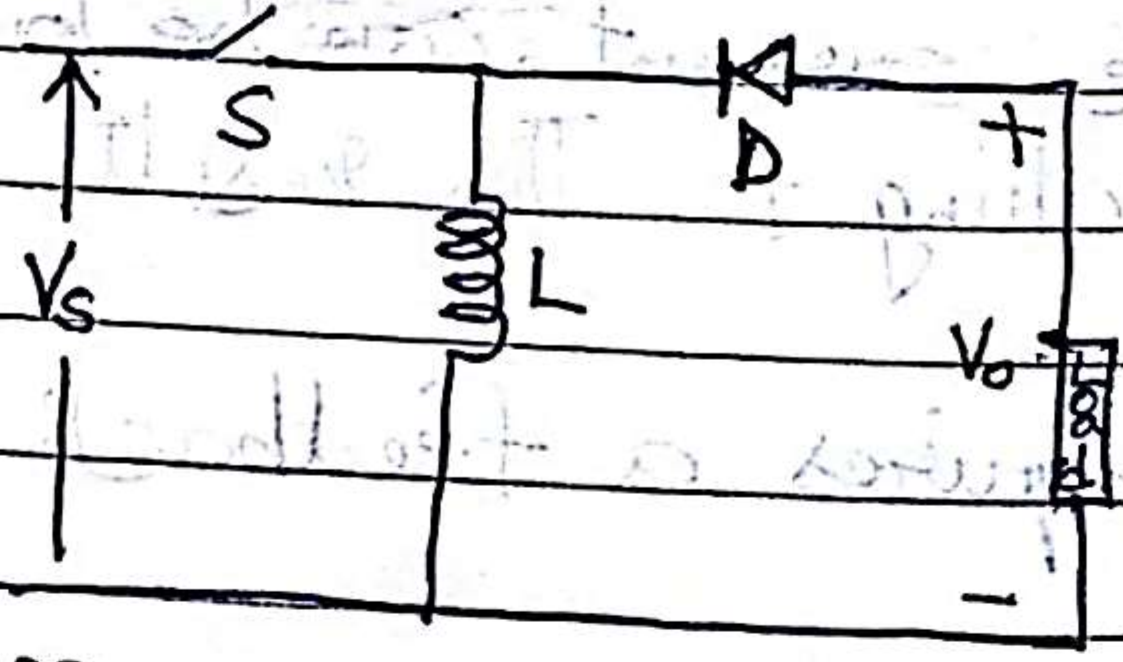
$$\Rightarrow V_s T = V_o T_{off}$$

$$\Rightarrow V_o = \frac{T}{T_{off}} V_s$$

$$\Rightarrow V_o = \frac{T}{T - T_{on}} V_s$$

$$\Rightarrow V_o = \frac{1}{1 - \alpha} V_s$$

BUCK-BOOST CHOPPER



When 'S' is ON

$$v_L = V_S$$

When 'S' is OFF

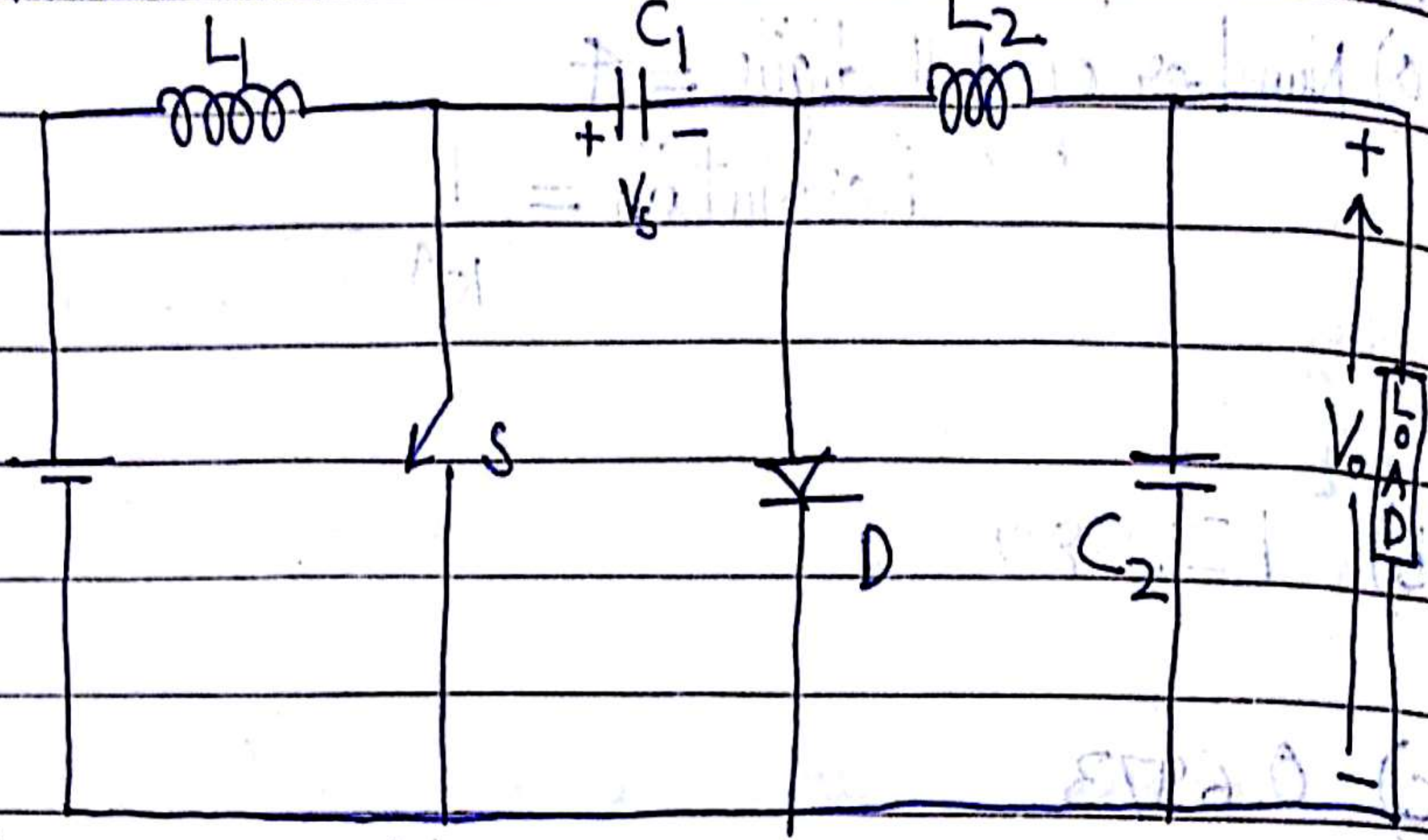
$$v_L = -V_0$$

$$V_S T_{on} = -V_0 T_{off}$$

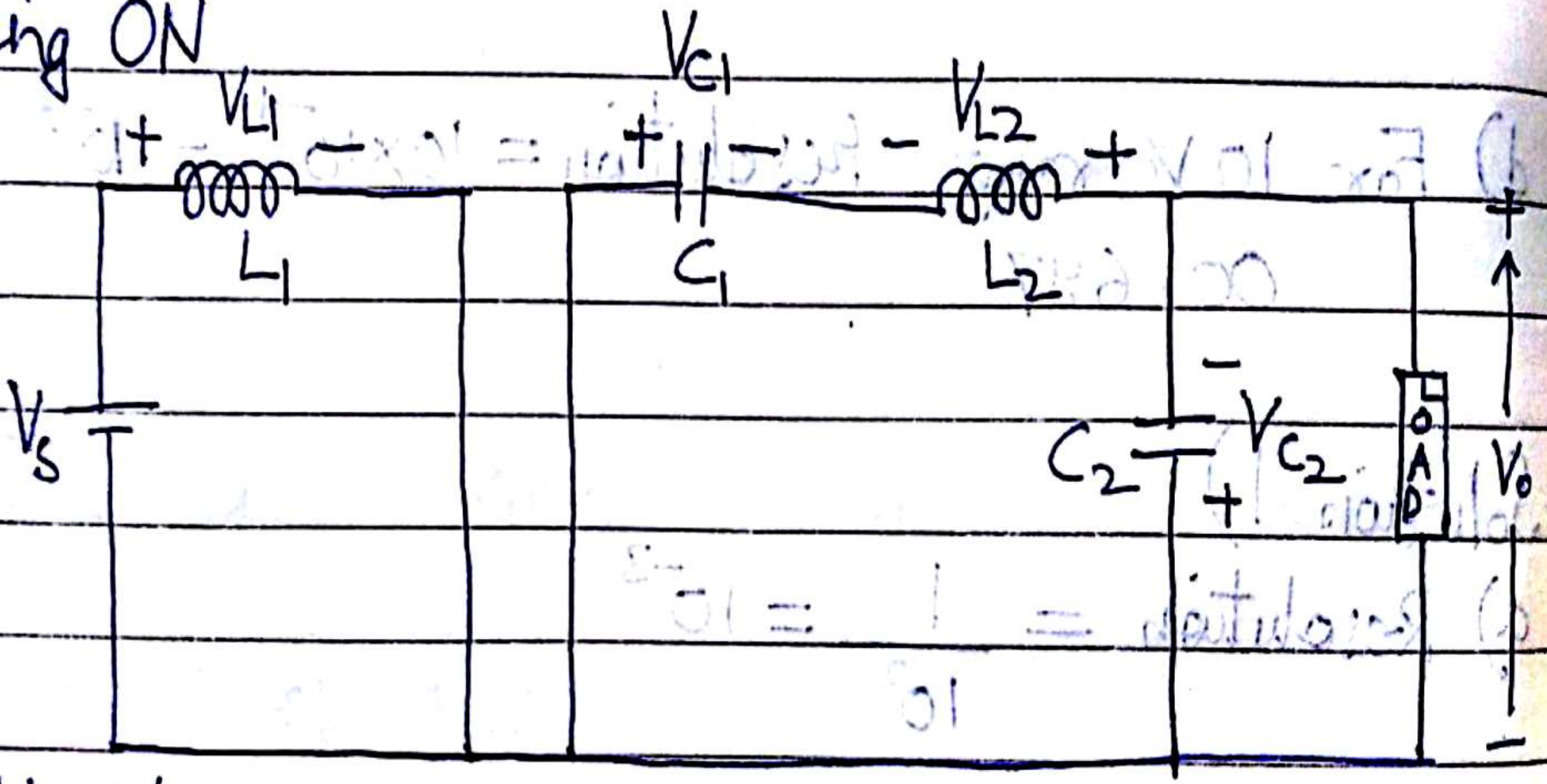
$$\Rightarrow V_0 = -V_S \frac{T_{on}}{T - T_{on}}$$

$$= -V_S \frac{\alpha}{1-\alpha}$$

CUK CONVERTER

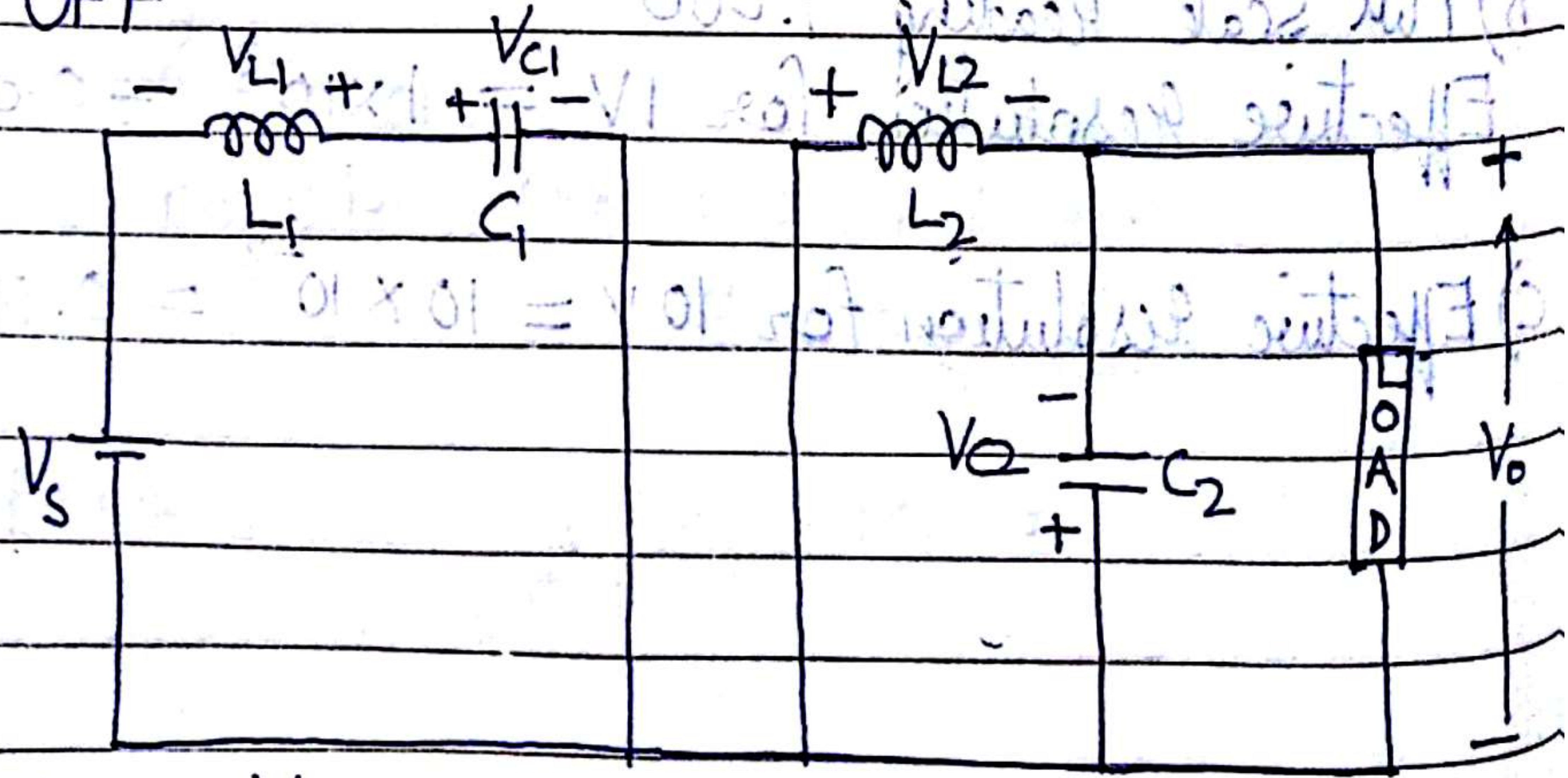


During ON



$V_S = V_{L1}$ $V_{L2} = V_{C1} + V_0$

During OFF



$$V_{L1} = V_{C1} - V_S \qquad V_{L2} = -V_0$$

For L_1

$$V_s t_{on} = (V_{C1} - V_s) t_{off}$$

$$\Rightarrow V_s T = V_{C1} t_{off}$$

$$\Rightarrow V_{C1} = \frac{V_s T}{t_{off}}$$

For L_2

$$(V_{C1} + V_o) t_{on} = -V_o t_{off}$$

$$V_{C1} t_{on} + V_o T = 0$$

$$\Rightarrow \frac{V_s T}{t_{off}} t_{on} + V_o T = 0$$

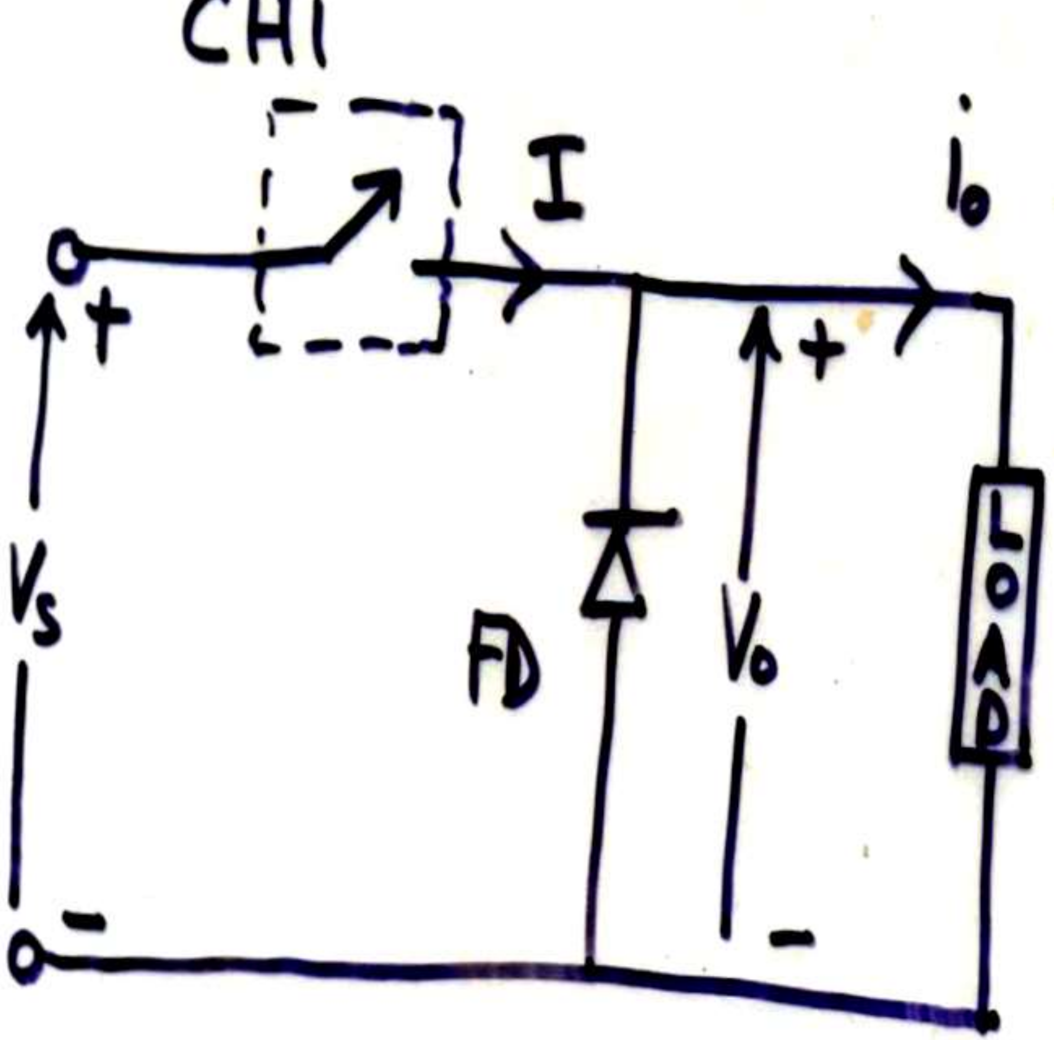
$$\Rightarrow V_s \frac{t_{on}}{t_{off}} + V_o = 0$$

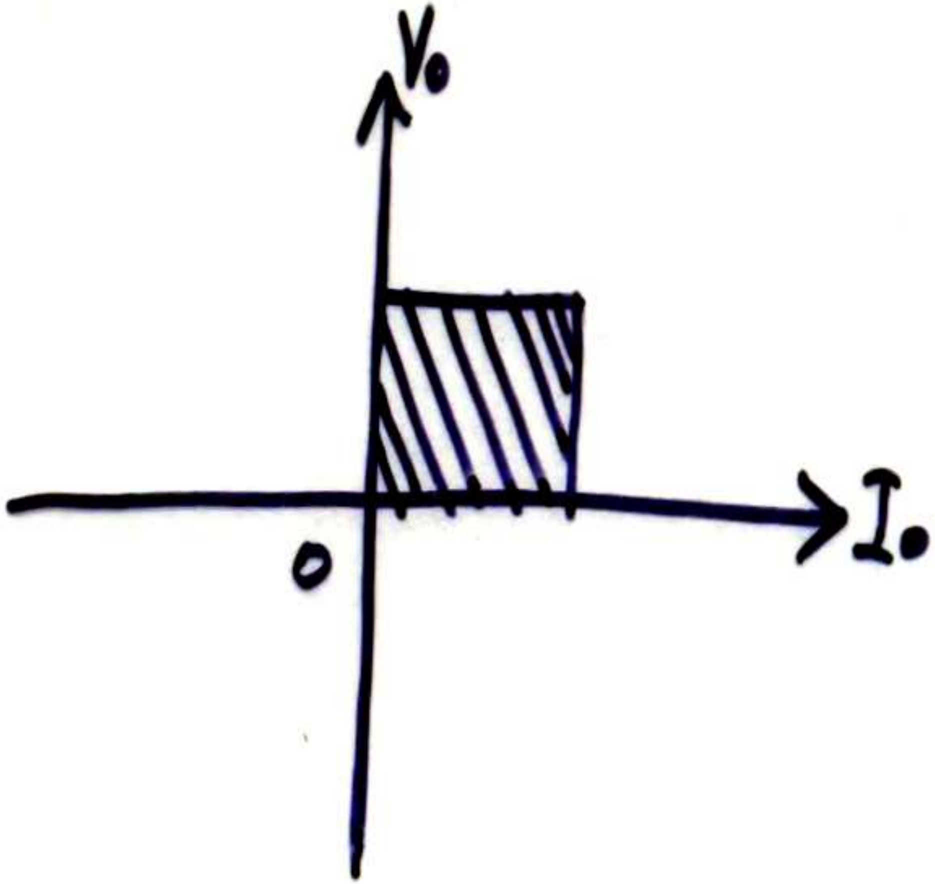
$$\Rightarrow -V_s \frac{(t_{on})/T}{(T - t_{on})/T} = V_o$$

$$\Rightarrow V_o = -\left(\frac{D}{1-D}\right) V_s$$

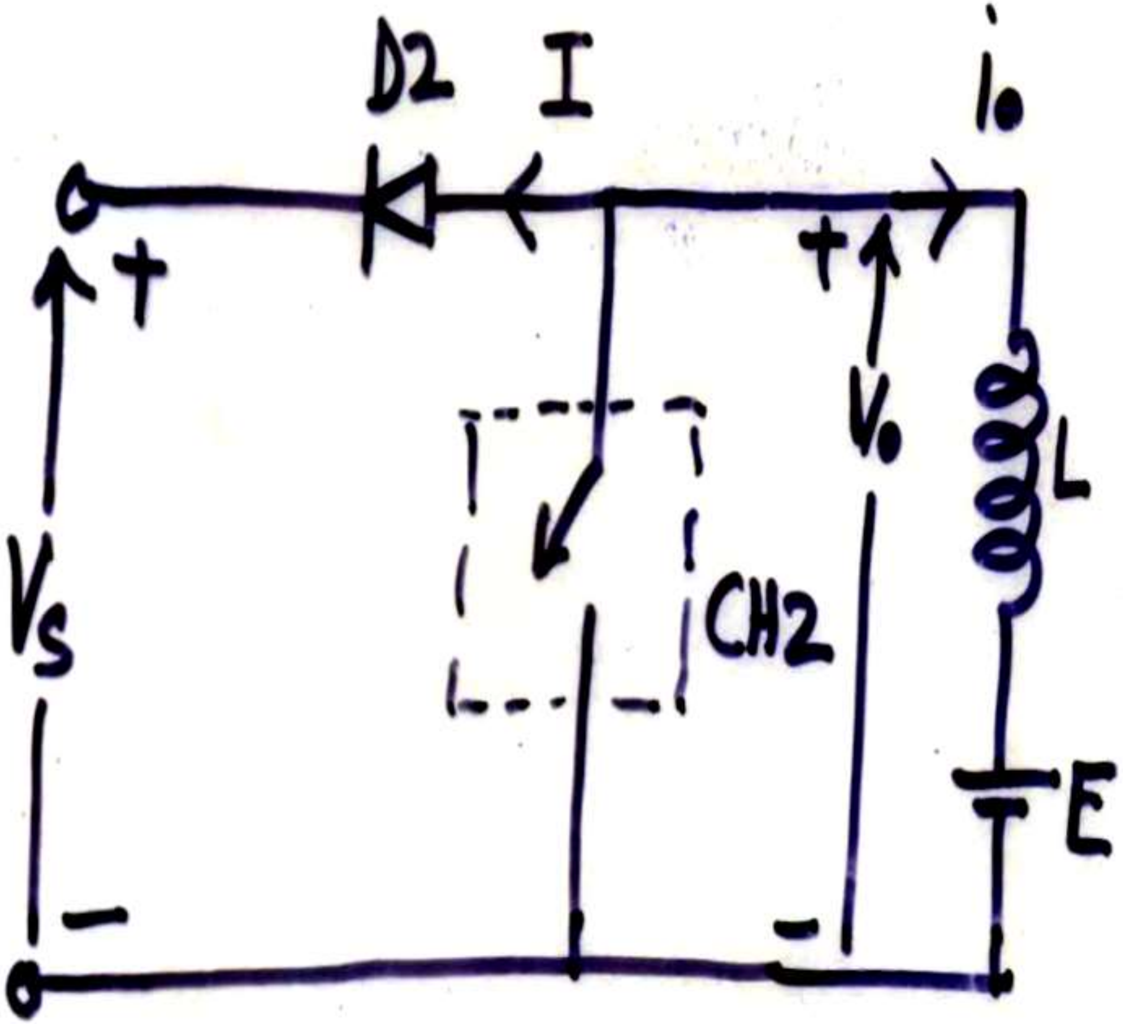
Quadrant based Classification

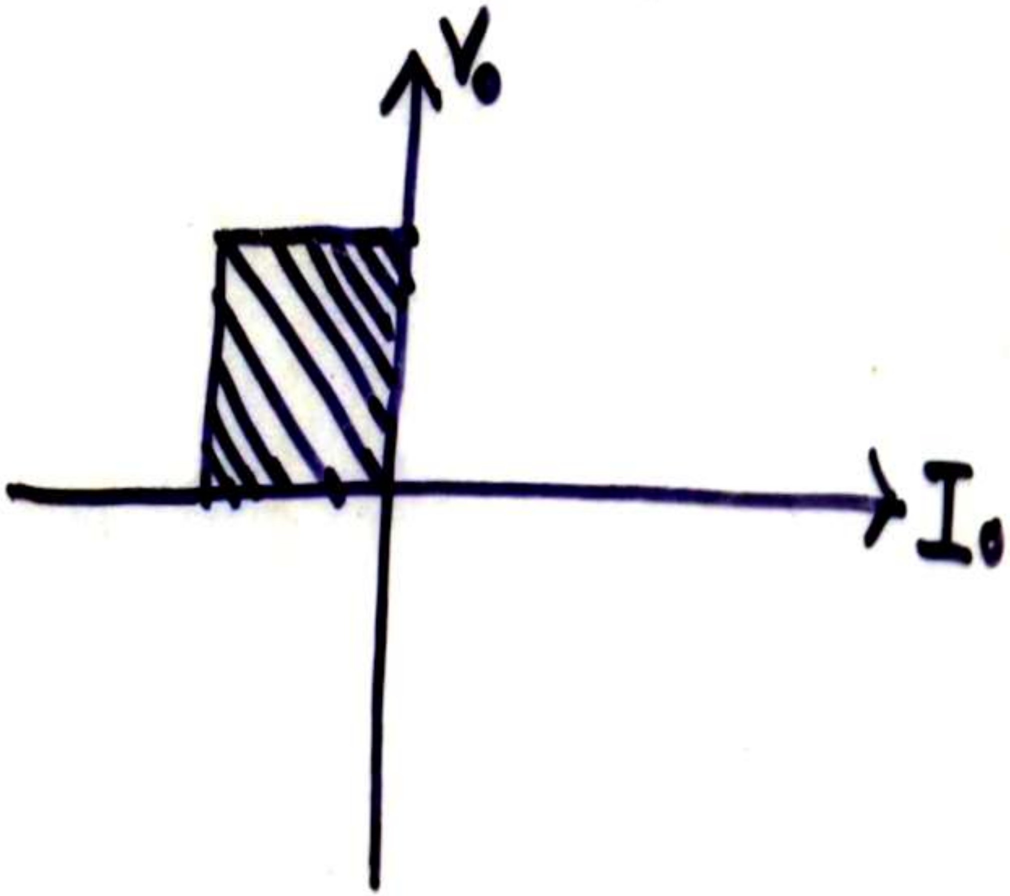
(1) I Quad/Type A



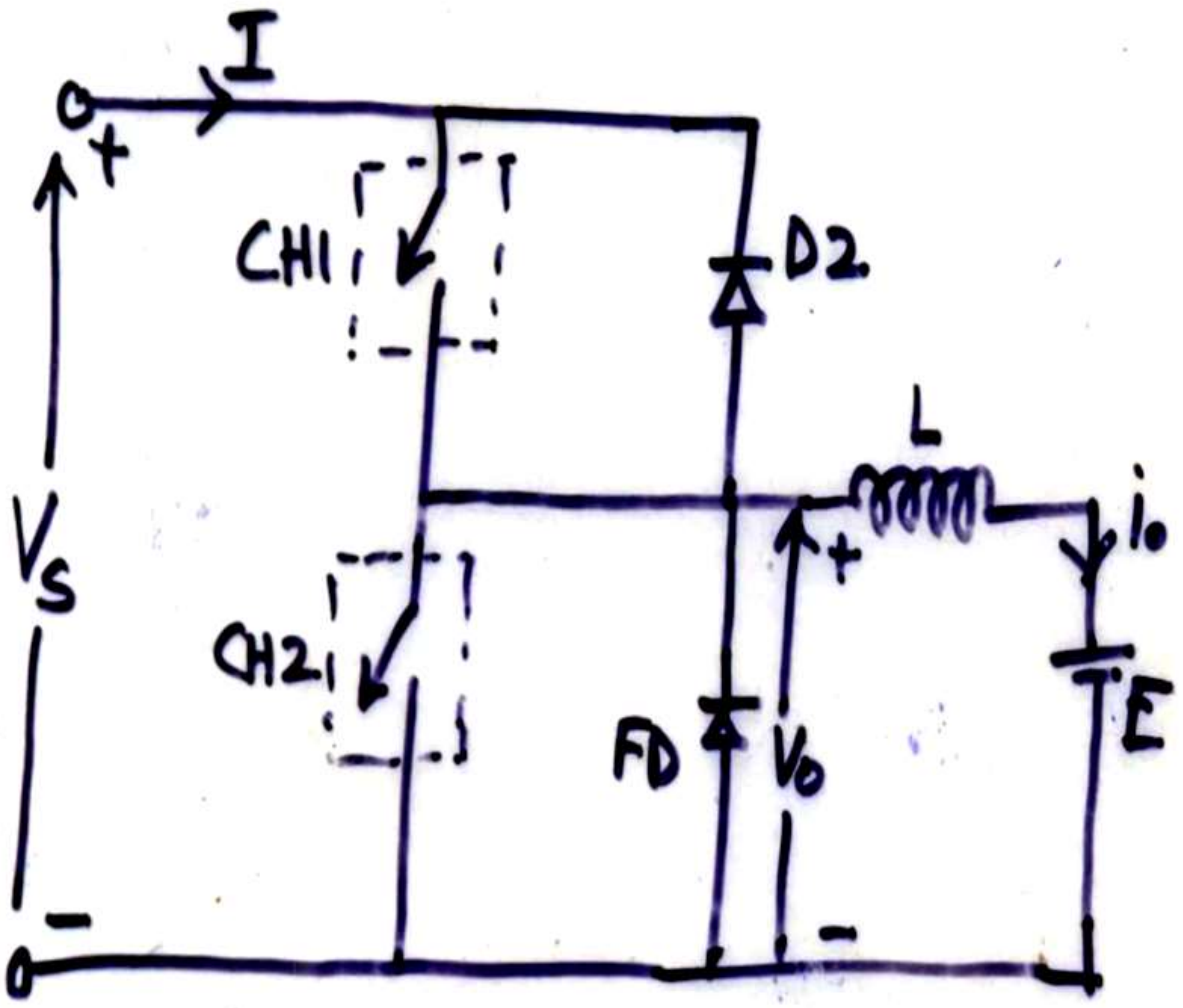


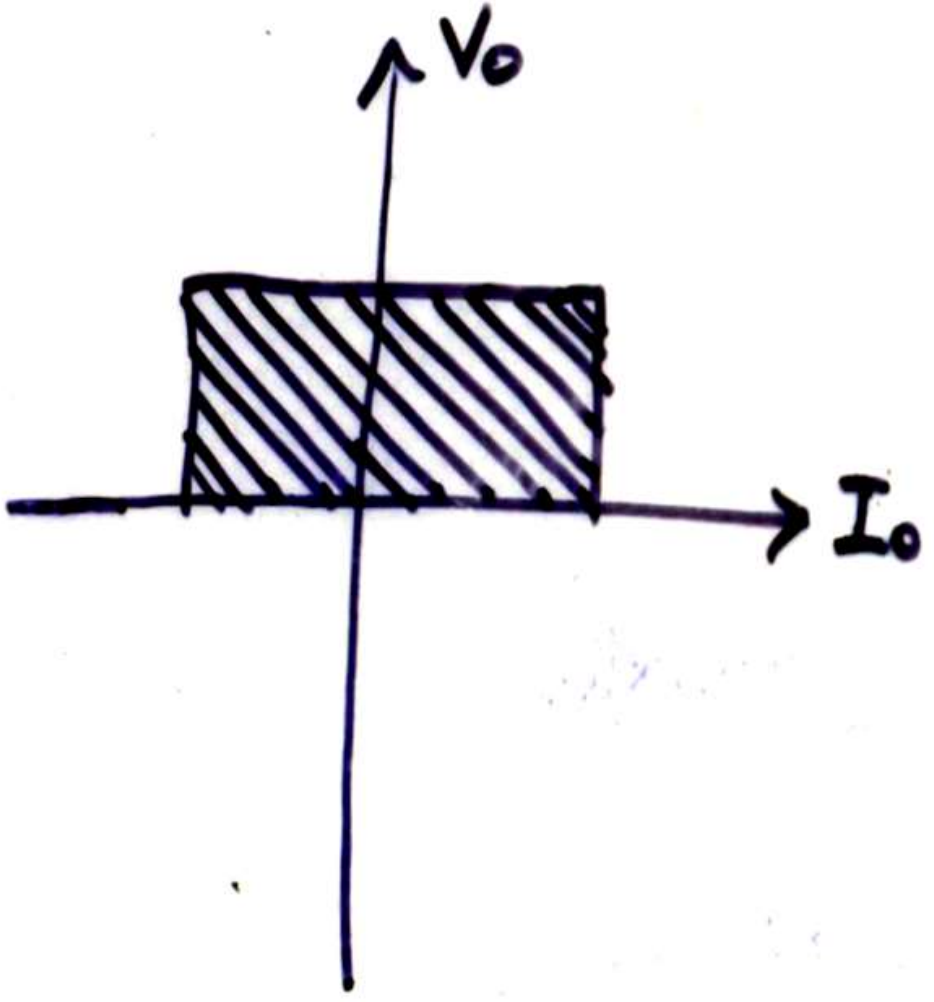
(2) II Quad/Type B



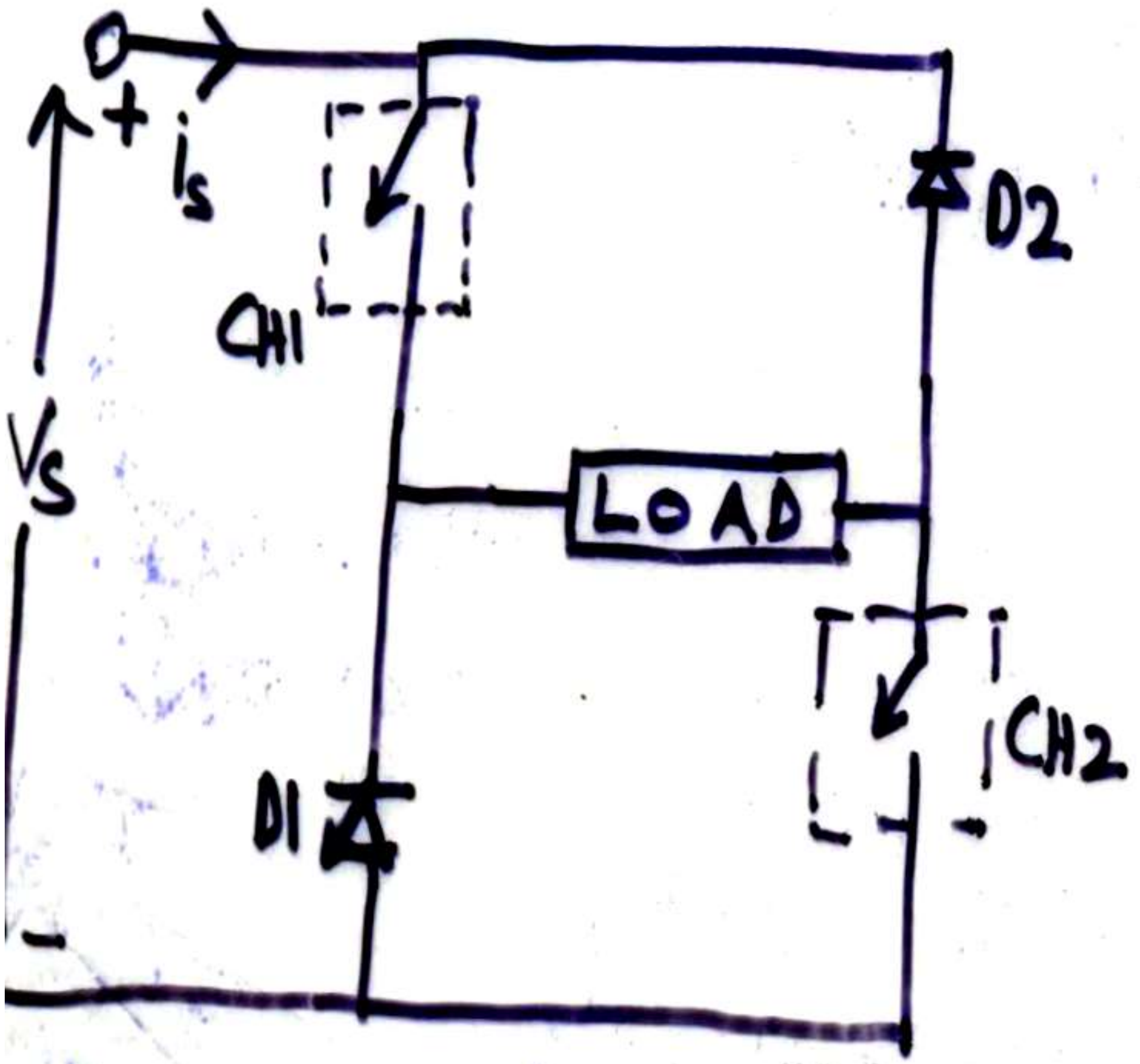


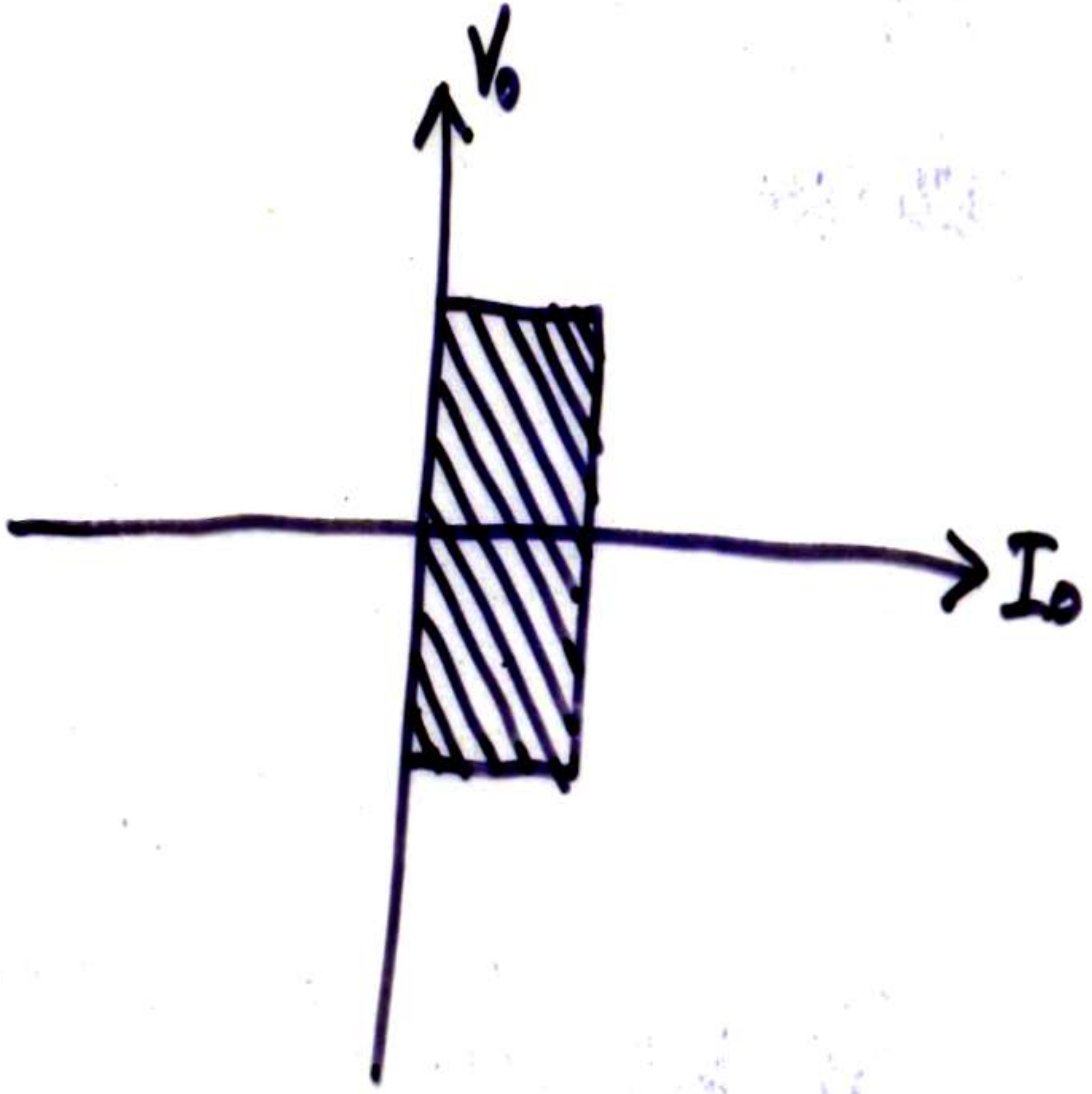
3) 2 Quad Type A/C



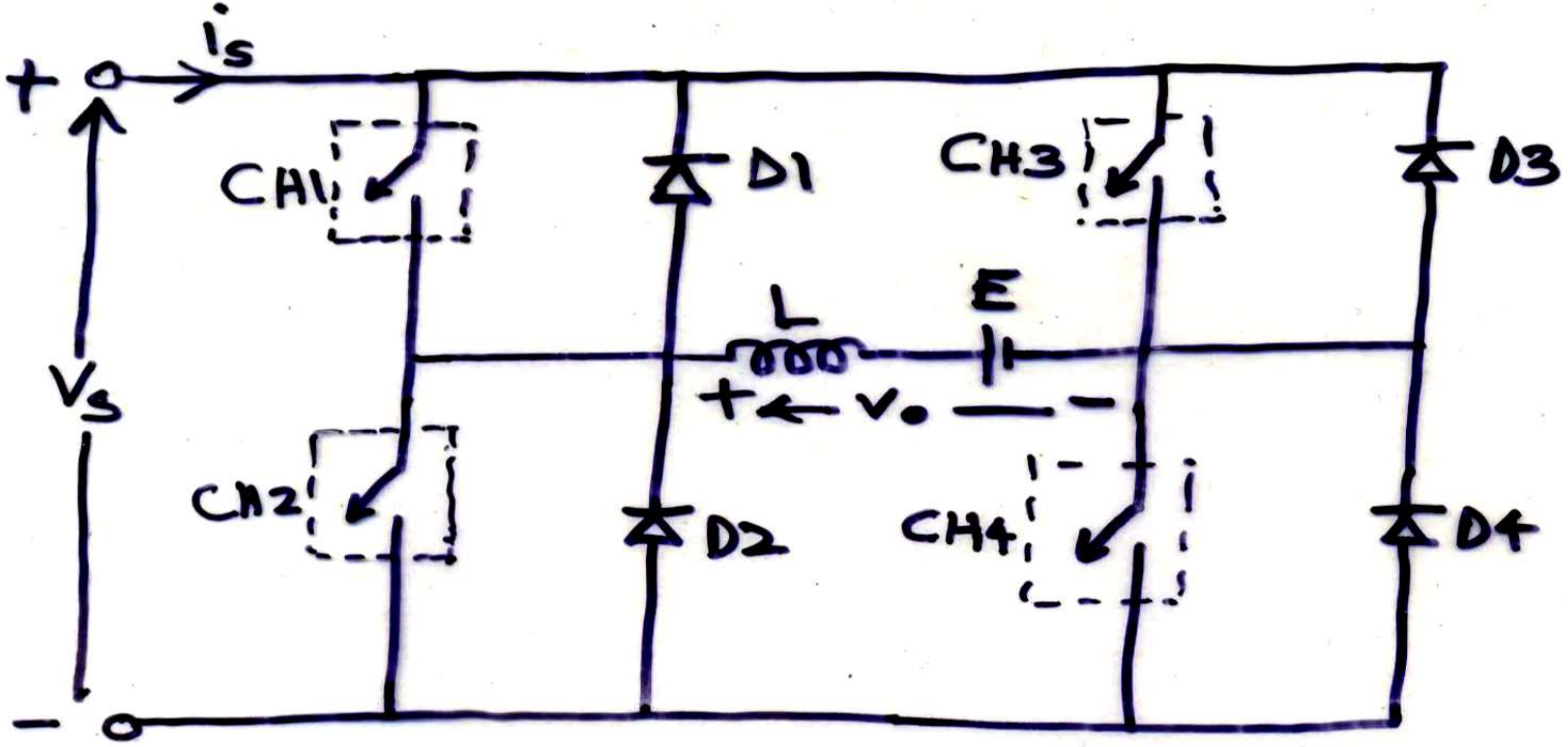


2. Quad Type B/D





(5) 4 Quad Type E



(5) 4 Quad Type E